

Federated Learning for Rural Clinics: Training Together Without Sharing Patient Records

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ABSTRACT

Small clinics collect valuable data but cannot share it, and each one alone is too small to train a reliable model. We simulate a federation of 24 clinics using de-identified screening records and compare federated averaging, a personalized variant and local training for predicting missed follow-up visits. The personalized federation reaches an AUROC of 0.83, close to a centralized upper bound of 0.85, while the median clinic training alone reaches only 0.71.

Keywords: federated learning, healthcare, privacy, non-IID data, tabular models

1. Introduction

A patient who misses a follow-up is easy to spot in hindsight and hard to predict in advance. A clinic with a few hundred records does not have enough examples to learn the pattern.

Federated learning lets clinics train a shared model while the records stay on their own machines [1]. The catch is that clinics differ a lot, which is where naive averaging struggles.

2. Method

We split de-identified records by clinic and keep each clinic's size and label balance as they are, so the federation is realistically uneven. Models are gradient boosted trees for the local baseline and small neural networks for the federated runs.

The personalized variant keeps a shared body and a small clinic-specific head, fine-tuned locally after each round [3]. Communication is limited to 40 rounds to match a weekly sync schedule.



Figure 1. Validation AUROC per round for twelve representative clinics. Dots mark the final round.

3. Results

Personalization matters most for the smallest clinics, which gain up to 0.15 AUROC over local training (Table 1). Plain averaging helps the median clinic but hurts two clinics with unusual patient populations.

Approach	AUROC median clinic	AUROC smallest	Data shared
Local only	0.71	0.62	none
Federated averaging	0.79	0.70	weights
Personalized (ours)	0.83	0.77	weights

Table 1. Missed follow-up prediction across the simulated federation.

4. Discussion

The honest limitation is that this is a simulation. A real deployment has flaky internet, different record formats and staff time. We describe a lightweight sync tool in Appendix A.

5. Conclusion

Clinics can get most of the benefit of pooled data without pooling it, as long as the model is allowed to be a little different at each site.

References

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